

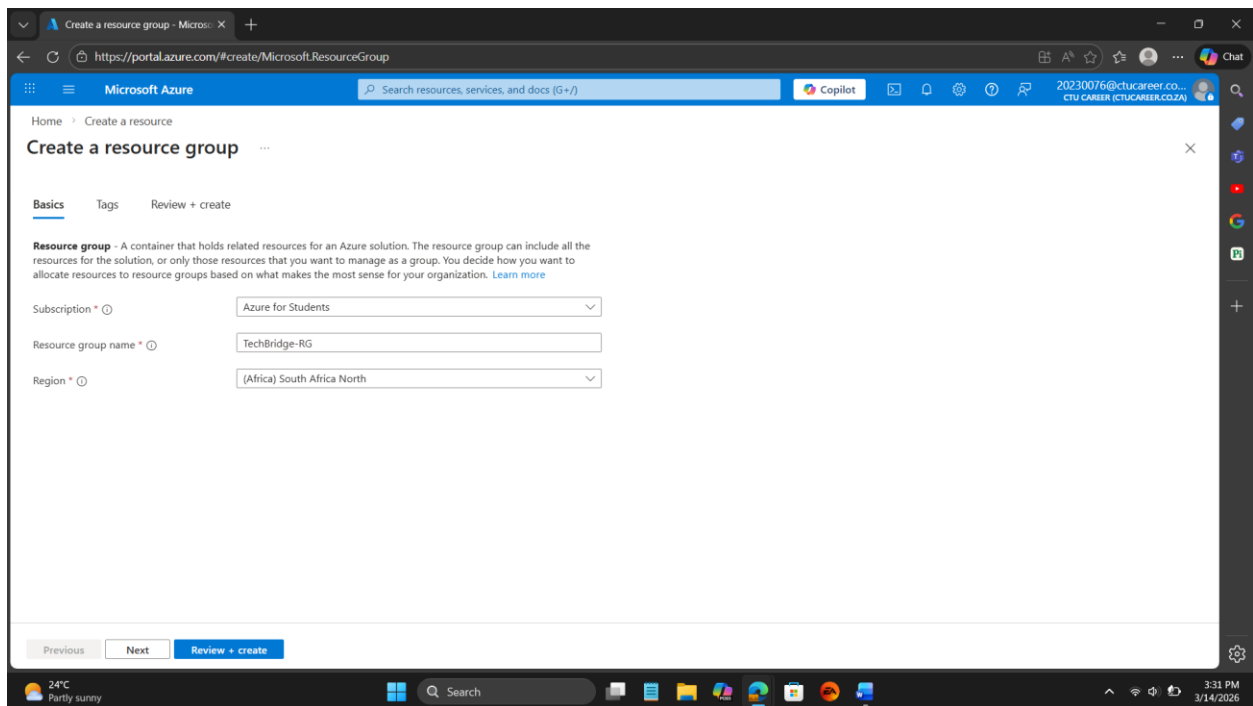
ONNEILE TLOUANA. 20230076. NET731. WEEK 2. NETWORK DESIGN

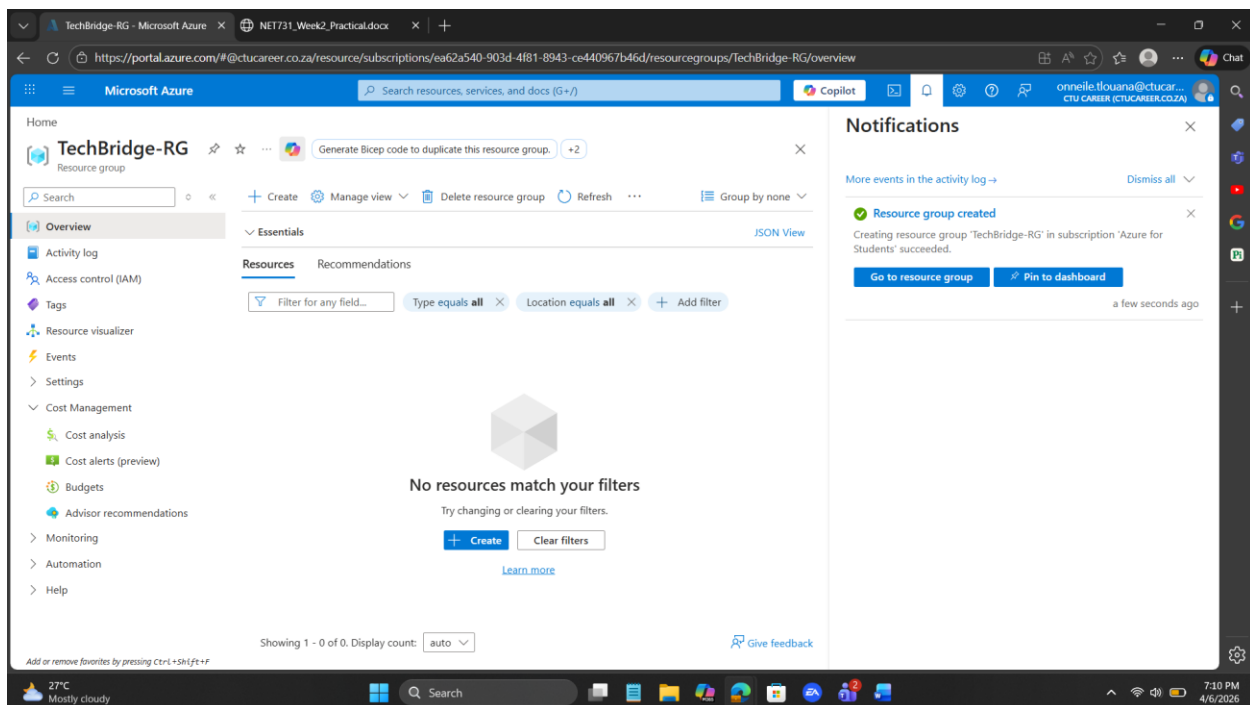
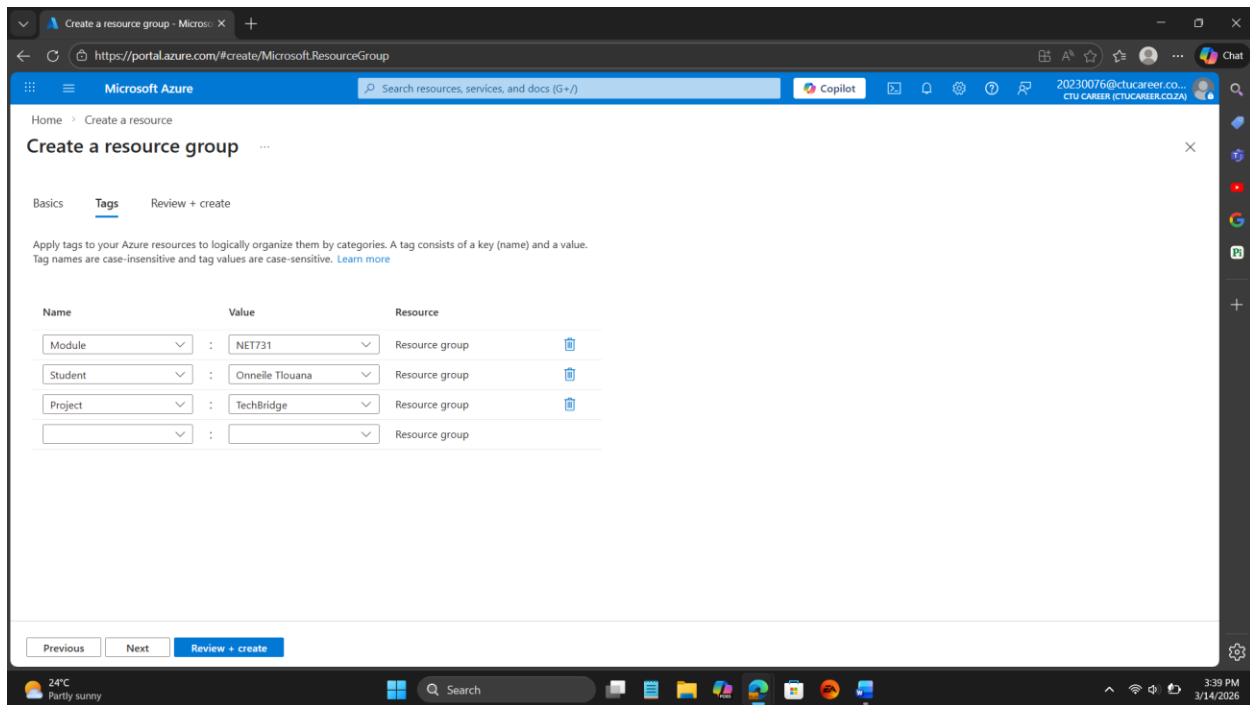
This practical focused on designing and configuring Azure networking components including Virtual Networks, subnets and Network Security Groups to support secure cloud communication.

Separate subnets were configured to improve network organization and security segmentation within the Azure environment.

Below are screenshots of

- Virtual Network created
- Configured address space and subnets
- Attachment of Network Security Group to subnets





Create virtual network - Microsoft

https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure

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Home

Create virtual network

BasicsSecurityAddress spaceTagsReview + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
Azure for Students

Resource group *
TechBridge-RG
[Create new](#)

Instance details

Virtual network name *
TechBridge-vnet

Region *
(Africa) South Africa North
[Deploy to an Azure Extended Zone](#)

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Create virtual network - Microsoft

https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM

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Create virtual network

BasicsSecurityAddress spaceTagsReview + create

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

☐ Allocate using IP address pools. [Learn more](#)

+ Add a subnet

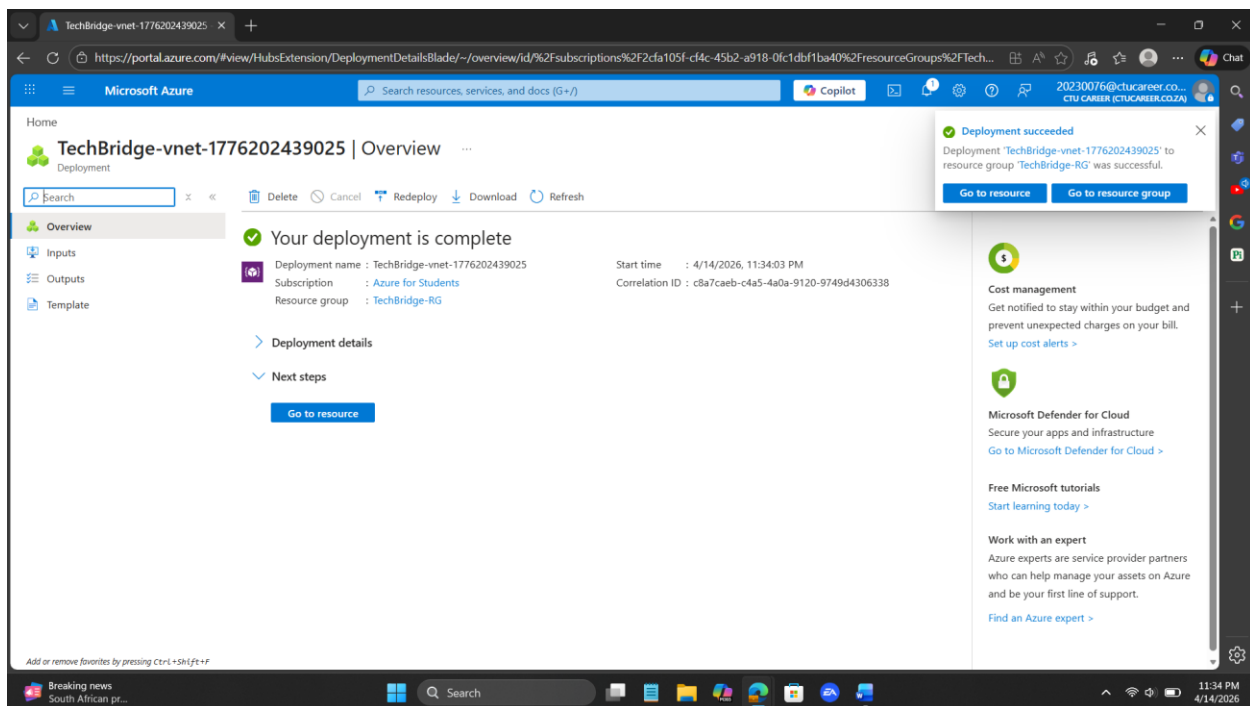
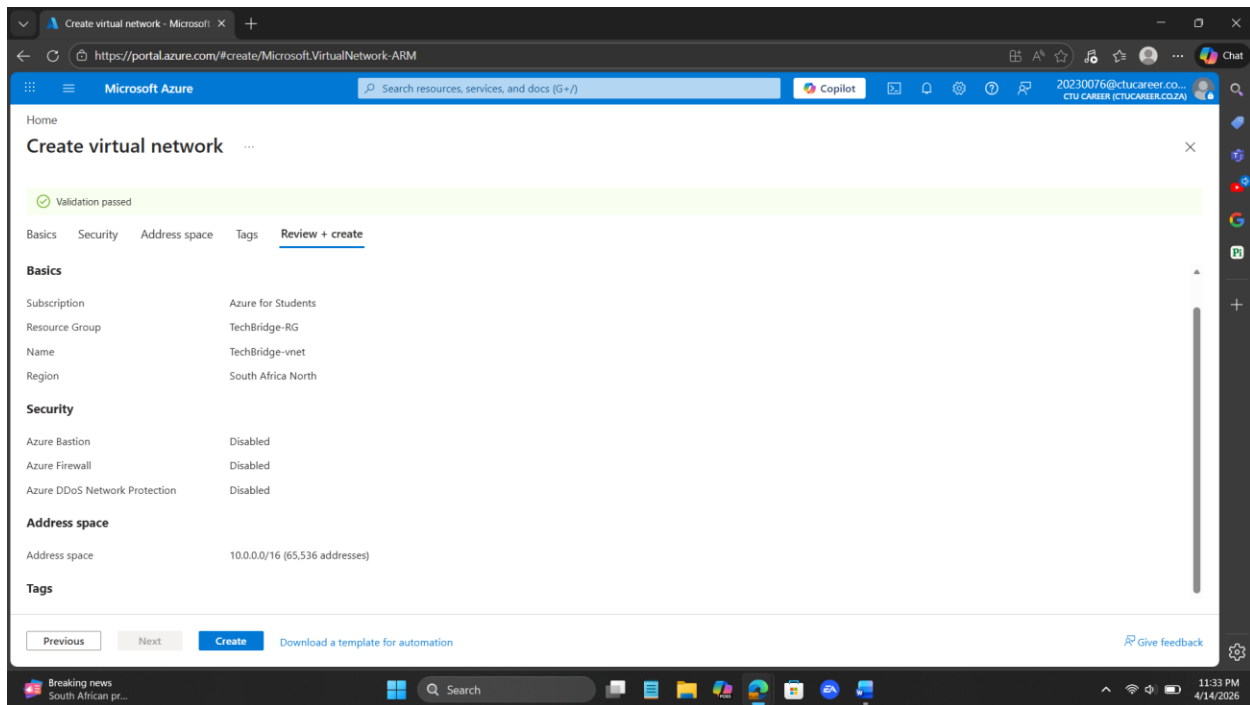
10.0.0.0/16
10.0.0.0 /16
10.0.0.0 - 10.0.255.255 65,536 addresses
Delete address space

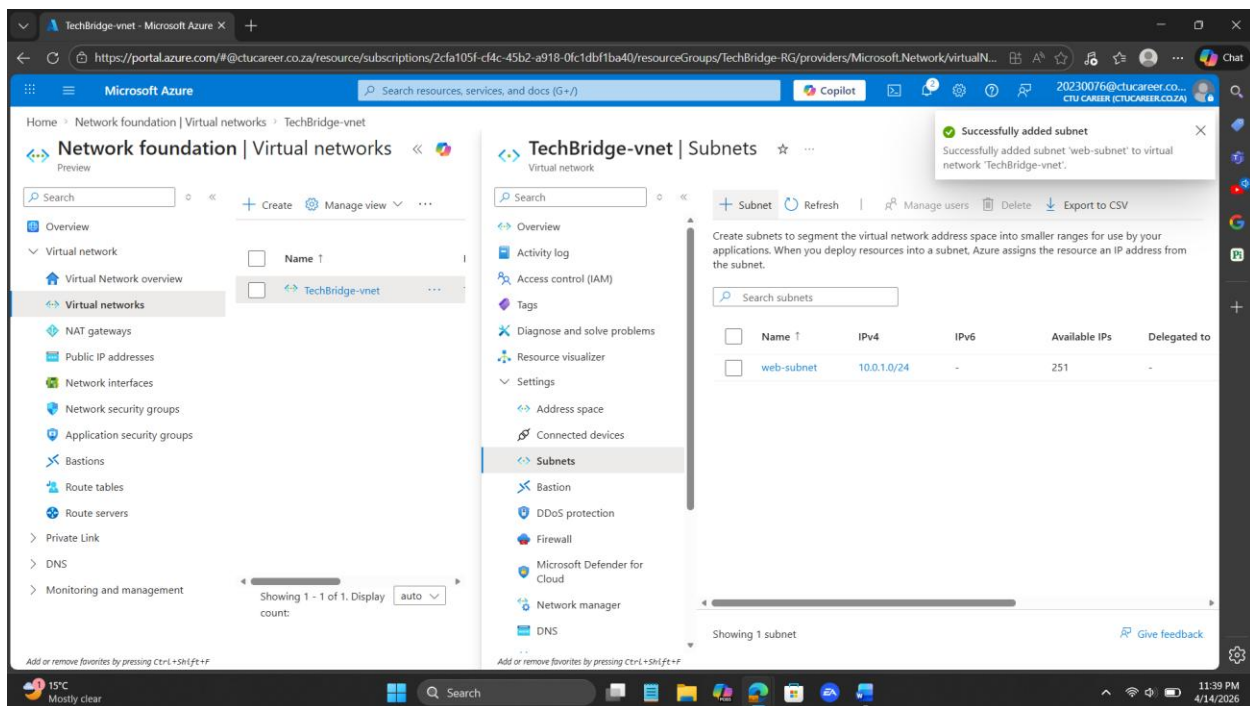
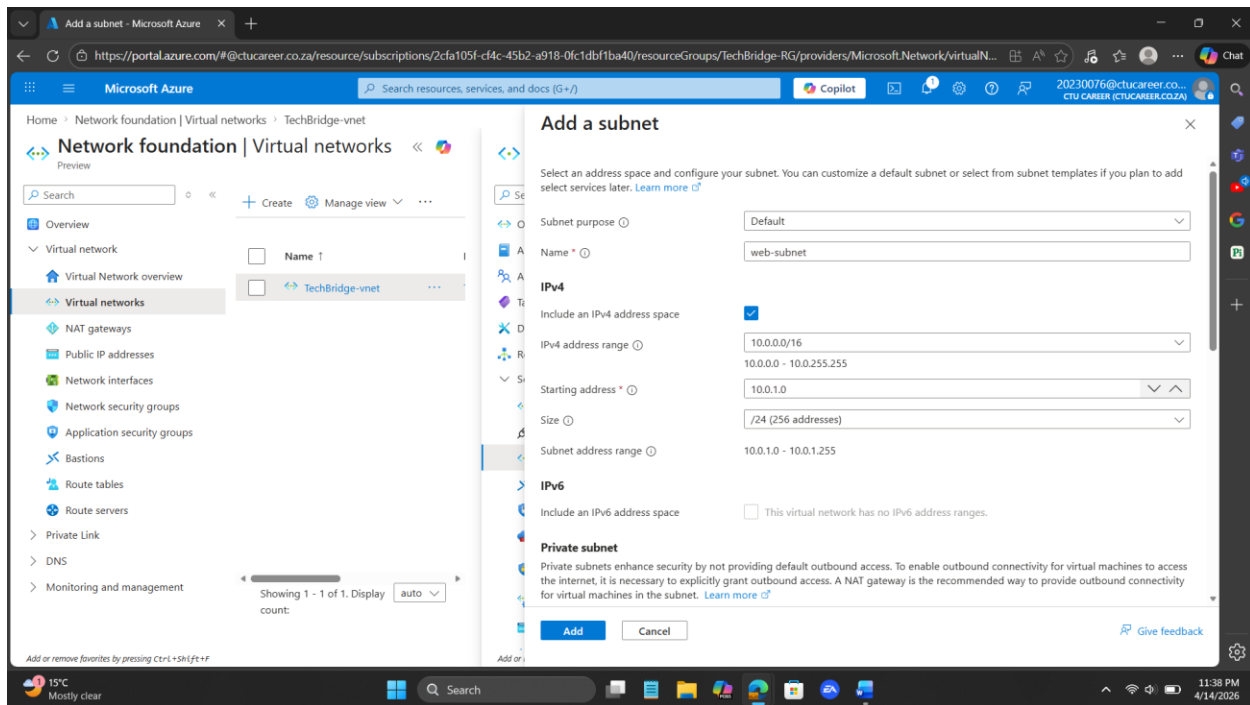
Subnets	IP address range	Size	NAT gateway
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Add IPv4 address space

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Microsoft Azure

Home > Network foundation > Virtual networks > TechBridge-vnet

Network foundation | Virtual networks

Overview

- Virtual network
 - Virtual Network overview
 - Public IP addresses
 - Network interfaces
 - Network security groups
 - Application security groups
 - Bastions
 - Route tables
 - Route servers
- Private Link
- DNS
- Monitoring and management

Showing 1 - 1 of 1. Display count: auto

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose: Default

Name: app-subnet

IPv4

Include an IPv4 address space: ☒

IPv4 address range: 10.0.0.0/16

Starting address: 10.0.0.0

Size: /24 (256 addresses)

Subnet address range: 10.0.0.0 - 10.0.2.255

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

[Add](#) [Cancel](#) [Give feedback](#)

Microsoft Azure

Home > Network foundation > Virtual networks > TechBridge-vnet

Network foundation | Virtual networks

Overview

- Virtual network
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- DNS
- Monitoring and management

Showing 1 - 1 of 1. Display count: auto

TechBridge-vnet | Subnets

Virtual network

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

- Address space
- Connected devices
- Subnets**
- Bastion
- DDoS protection
- Firewall
- Microsoft Defender for Cloud
- Network manager
- DNS

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name	IPv4	IPv6	Available IPs	Delegated to
web-subnet	10.0.1.0/24	-	251	-
app-subnet	10.0.2.0/24	-	251	-

Showing 2 subnets

[Give feedback](#)

Create network security group - M X +

https://portal.azure.com/#view/Microsoft_Azure_Network/CreateNetworkSecurityGroupBladeV2/_provisioningContext~/%7B"initialValues"%3A%7B"subscriptionId"%3A%5B"2cfa...

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Home > Network foundation | Network security groups

Create network security group

Basics Tags Review + create

Project details

Subscription * Azure for Students

Resource group * TechBridge-RG
[Create new](#)

Instance details

Name * TechBridge-nsg

Region * South Africa North

[Review + create](#) < Previous Next: Tags > [Download a template for automation](#)

15°C Mostly clear 11:46 PM 4/14/2026

CreateNetworkSecurityGroupBladeV2 - M X +

https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/_overview/id/%2Fsubscriptions%2F2cfa105f-cf4c-45b2-a918-0f1dbf1ba40%2FresourceGroups%2FTech...

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Home

CreateNetworkSecurityGroupBladeV2-20260414234534 | Overview

Deployment

Search x < Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : CreateNetworkSecurityGroupBladeV2-202604142... Start time : 4/14/2026, 11:46:44 PM

Subscription : Azure for Students Correlation ID : f43cda78-b98a-480c-92c6-a38cf8bd9d53

Resource group : TechBridge-RG

> Deployment details

> Next steps

[Go to resource](#)

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Microsoft Azure portal interface showing the "Add inbound security rule" dialog for the "TechBridge-nsg" network security group. The dialog is configured with the following settings:

- Destination: Any
- Service: Custom
- Destination port ranges: 80
- Protocol: TCP
- Action: Allow
- Priority: 100
- Name: Allow-HTTP

The background shows the "Inbound security rules" list for "TechBridge-nsg". The table below represents the data visible in the background interface:

Priority	Name	Port	Protocol
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancer...	Any	Any
65500	DenyAllInBound	Any	Any

Microsoft Azure portal interface showing the "Add inbound security rule" dialog for the "TechBridge-nsg" network security group. The dialog is configured with the following settings:

- Destination: Any
- Service: Custom
- Destination port ranges: 443
- Protocol: TCP
- Action: Allow
- Priority: 110
- Name: Allow-HTTPS

The background shows the "Inbound security rules" list for "TechBridge-nsg". The table below represents the data visible in the background interface:

Priority	Name	Port	Protocol
100	Allow-HTTP	80	TCP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancer...	Any	Any
65500	DenyAllInBound	Any	Any

Microsoft Azure portal interface showing the "Add inbound security rule" dialog for the "TechBridge-nsg" network security group.

Dialog Fields:

- Destination: Any
- Service: Custom
- Destination port ranges: 1433
- Protocol: TCP
- Action: Allow
- Priority: 120
- Name: Allow-App-Internal

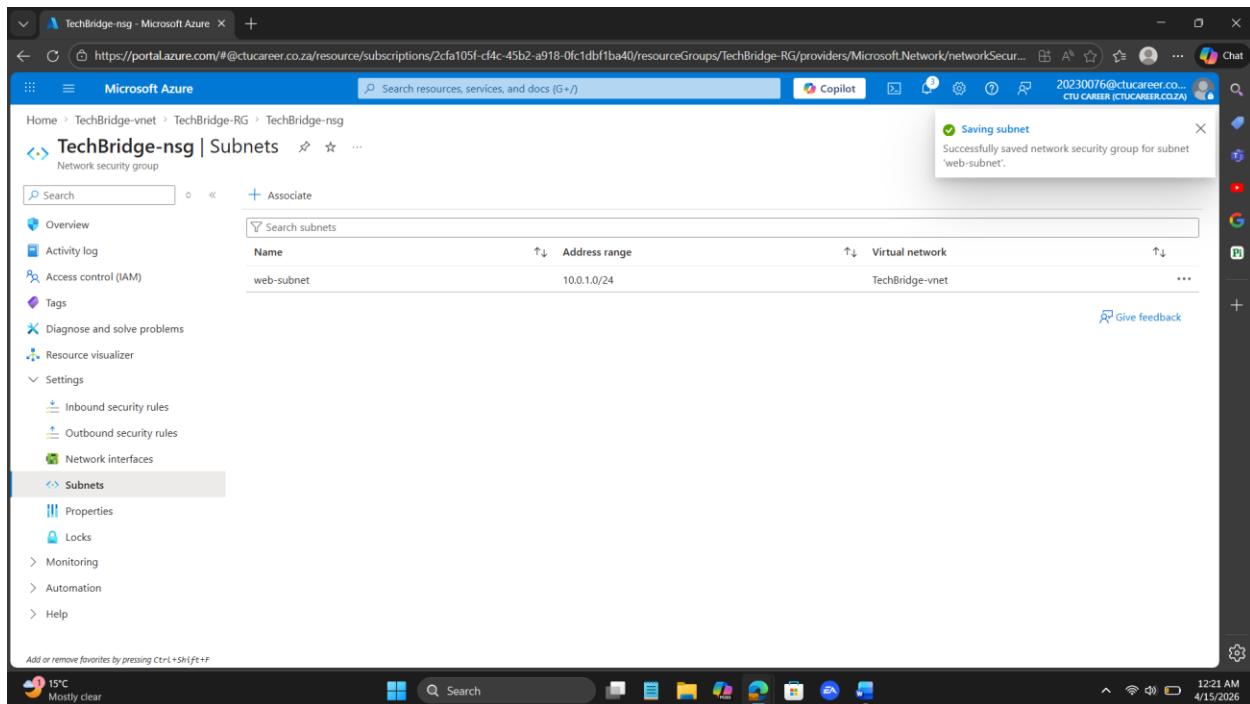
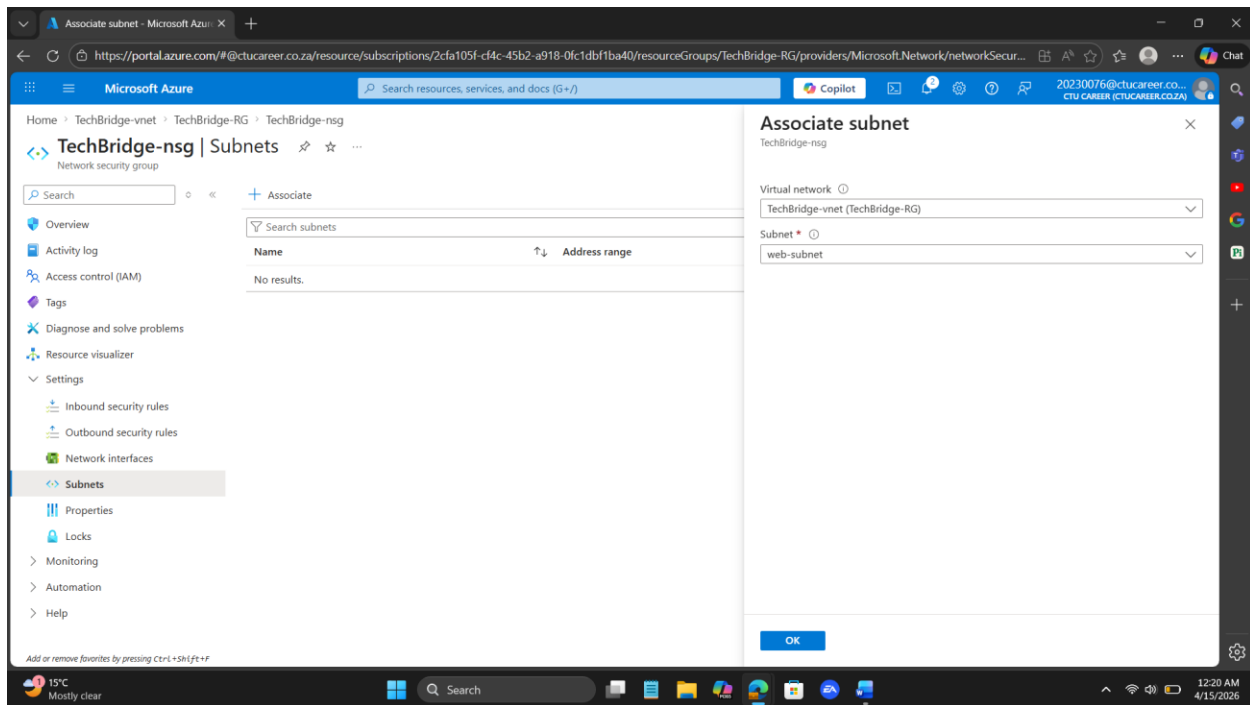
Background Table: Inbound security rules

Priority	Name	Port	Protocol
100	Allow-HTTP	80	TCP
110	Allow-HTTPS	443	Any
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancer	Any	Any
65500	DenyAllInBound	Any	Any

Microsoft Azure portal interface showing the "TechBridge-nsg" network security group with the "Allow-App-Internal" rule added.

Background Table: Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-HTTP	80	TCP	Any	Any	Allow
110	Allow-HTTPS	443	Any	Any	Any	Allow
120	Allow-App-Internal	1433	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancer	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny



REFLECTION

This practical improved my understanding of Azure networking, subnetting, and traffic control using Network Security Groups.

SKILLS DEMONSTRATED

- Azure Networking
- Virtual Networks
- Subnet Configuration
- Network Security Groups
- Traffic Segmentation